

FUNDAMENTALS OF DISTRIBUTED AND CLOUD COMPUTING (21B12CS313)

DISTRIBUTED SYSTEM???

Computer Networks

An interconnected collection of autonomous computers

What do we mean by autonomous here?

If one computer can forcibly start, stop, or control another one, then the computers are not autonomous.

Computer Networks contd.

Users must explicitly log onto one m/c, explicitly submit jobs remotely, explicitly move files around and generally handle all the network management personally.

Distributed Systems

- Existence of multiple computers is transparent to the user.
- Allocation of jobs to processes and files to disks, movement of files between where they are stored and where they are needed, and all other system functions must be automatic.
- Is a software built on top of a network

What is the difference between
computer networks and Distributed
Systems?

The distinction between a network and distributed system lies with the s/w (especially the OS), rather than with the hardware.

- A distributed system is one in which components located at networked computers communicate and coordinate their actions only by **passing messages**.

In other words, communication is only by **message passing**.

Characteristics of DS

- Concurrency of components: Coordination of concurrently running programs that share resources.
- Lack of Global Clock: There is no single notion of the correct time(This is the consequence of the fact that only communication is by sending messages through a network.
- Independent failures of Components

Examples of DS: The Internet, Intranet, Mobile computing etc.

Motivation for Constructing and using DS

- Sharing of resources is a main motivation for constructing DS
- Resources:
 - Hardware resources: Disks, printers etc.
 - Software resources: Files, Databases & data objects of all kinds including stream of video frames, audio connections and so on.

- Sharing of higher level resources is of greater significance

- Sharing data (shared database)

- Set of web pages

- Search Engine

Sharing disks & processors are less bothered.

Definition of a Distributed System

A distributed system is:

- A collection of independent computers that appears to its users as a single coherent system.

A Distributed System (DS) consists of a collection of autonomous computers linked by a computer network and equipped with DS software.

DS software enables computers to coordinate their activities and to share the resources of the system - hardware, software and data.

Users of a DS should perceive a single, integrated computing facility even though it may be implemented by many computers in different locations.

Key Challenges in Design and Implementation

- The following Implementation Challenges must be met to ensure the usefulness of distributed systems:
 - Heterogeneity
 - Resource Sharing
 - Openness
 - Concurrency
 - Scalability
 - Failure Handling (Fault tolerance)
 - Transparency
 - Security

System and application software must be carefully designed to ensure these challenges are met.

Heterogeneity

- Heterogeneity - variety and difference
- A DS should enable users to access services and run applications over a heterogeneous collection of computers and networks
- Heterogeneity applies to all of the following:
 - Networks
 - Computer hardware - PC and Apple architecture
 - Operating systems - Unix, Windows 2000
 - Programming languages and Databases - Java, C++, Oracle, SQL Server
 - Implementations by different developers

Heterogeneity

- Networks
 - The Internet consists of many different types of networks (Ethernet, Token Ring, ATM etc) their differences are
masked by the fact that
all of the computers attached to them use the TCP/IP suite of
protocols to communicate with
each other

Heterogeneity

- Computer hardware
 - Data types, eg integers, are represented in different ways on different hardware:
 - Byte ordering - Big endian, Little endian
 - Size - 16 bits, 32 bits, 64 bits
 - These differences must be dealt with if messages are to be exchanged between programs running on different hardware
- Operating Systems
 - Unix and Windows 2000 provide different APIs to interface to the TCP/IP Suite of protocols

Heterogeneity

- Programming Languages and Databases
 - Use different representations for characters and data structures - arrays and records
 - These differences must be addressed
- Implementations by different developers
 - Programs written by different developers cannot communicate with one another unless they use common standards
 - standards need to be agreed and adopted

Resource Sharing

- The term resource characterizes the range of things that can be shared usefully in a distributed system:
 - Hardware: disks, printers
 - Software : files, databases, compilers, applications
- Sharing reduces costs - as the cost of a resource decreases the need to share lessens - TVs 1956, Telephones 1960s
- Is necessary to support groups of users, working cooperatively (shared work environment) - project work

Resource Sharing

- Resources physically encapsulated within one of the computers in a DS can only be accessed from other computers by via a network
- Each set of resources of a particular type (printer) must be managed by a program (Resource Manager) that offers a communication interface that enables the resource to be:
 - Accessed
 - The provision of a Naming Scheme (inkptr1, laserptr1, colourptr1) for each type of resource, enabling individual resources to be accessed from any location
 - Mapping of resource names to communication addresses (each printer has an MAC and IP address)
 - Manipulated and updated
 - The synchronisation of concurrent accesses to ensure consistency, eg Resource = Database

Openness

- Determines whether the system can be extended and re-implemented in various ways at:
 - Hardware level, addition of computers to the network
 - Software level, introduction of new services and re-implementation of old ones

Openness

- Open systems are characterized by the fact that their key interfaces are published, thus making them available to software developers
- Open DS's are based on the provision of a uniform communication mechanism (eg RPC, CORBA RMI) and published interfaces for access to shared resources
- Open DS's can be constructed from heterogenous hardware and software, from different vendors. But the conformance of each component to the published standard must be carefully tested and verified if the system is to work correctly

Concurrency

- Concurrency and parallel execution arise in a DS due to:
 - the separate activities of users,
 - the independence of resources
 - and the location of processes in separate computers
- If there are M computers in a DS with one CPU each, then up to M processes can run in parallel (execute simultaneously)
- If several processes access the same resource concurrently their actions must be synchronised to ensure they do not conflict

Scalability

- DSs must operate effectively and efficiently at many different scales:
 - Two workstations and file server
 - LAN with several hundred workstations and many file servers
 - Several LANs connected to form a WAN
- System and application software should not need to change when the scale of the system increases
- The work (time) involved in processing any single request to access a shared resource should be independent of the size of the network

Scalability

- No single resource - hardware or software is assumed to be in restricted supply, as demand for a resource grows, it should be possible to extend the system to meet the demand, at reasonable cost

Transparency in a Distributed System

Transparency	Description
Access	Hide differences in data representation and how a resource is accessed
Location	Hide where a resource is located
Migration	Hide that a resource may move to another location
Relocation	Hide that a resource may be moved to another location while in use
Replication	Hide that a resource may be shared by several competitive users
Concurrency	Hide that a resource may be shared by several competitive users
Failure	Hide the failure and recovery of a resource
Persistence	Hide whether a (software) resource is in memory or on disk

Different forms of transparency in a distributed system.

Scalability Problems

Concept	Example
Centralized services	A single server for all users
Centralized data	A single on-line telephone book
Centralized algorithms	Doing routing based on complete information

Examples of scalability limitations.

Service

A distinct part of computer system that manages a collection of related resources and presents their functionality to users and applications.

e.g. file service- to read, write and delete Operations.

Effective sharing

Each resource must be managed by a program that offers a communication interface enabling the resource to be accessed and updated reliably and consistently.

Servers

- Refers to a running program (a process) on a networked computer that accepts requests from programs running on the computers to perform a service and respond appropriately.
- Requests and replies are nothing but messages.

- Client- requests, active, runs periodically
- Server- replies, passive, runs continuously
- By default, the terms clients and servers refer to processes rather than the computers.
- Same process may be both client & server
(since servers sometimes invoke operation on other servers)

System Models

- Two models:
 - **Architectural Model**: concerned with the placement of its part and the relationships between them.
 - Client-server Model
 - Peer-process Model
 - **Fundamental Model**: concerned with more formal description of the properties that are common in all of the architectural models.
 - Interaction Model: delays
 - Failure Model: from a variety of failures
 - Security Model: vulnerable to security attacks

- **Interaction Model**: deals with performance and with the difficulty of setting time limits in a DS. e.g. message delivery.
- **Failure Model**: attempts to give a precise specification of the faults that can be exhibited by processes and communication channels. It defines reliable communication and correct processes.
- **Security Model**: discusses the possible threats to processes and communication channels. It introduces the concepts of a secure channel, which is secure against those threats.

Difficulties for and threats to DS

□ Widely varying modes of use:

- Wide variations in workload
- Some parts of the systems may be disconnected
- Some applications have special requirement for high commn. b/w e.g. multimedia applications.

□ Wide range of system environments:

- Requires accomodation of hetrogeneous h/w, o/s and n/ws.
- n/ws may differ widely in performance. e.g. wireless
- Systems of widely differing scales must be supported e.g. ranging from tens of computers to millions of computers

□ Internal problems

- Non synchronized clocks
- Conflicting data updates
- Many modes of h/w & s/w failures

□ External Threat

- Attacks on data integrity and secrecy
- Denial of service